Model for "D3"

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General Condition

- Basis type: `lgs`
- SAMB selection:
  - Type: `[Q, G]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1, A_2, E]
  - Spin (s): `[0, 1]`
- Atomic selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1, A_2, E]
  - Spin (s): `[0, 1]`
- Site-cluster selection:
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1, A_2, E]
- Bond-cluster selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1, A_2, E]
- Max. neighbor: `10`
- Search cell range: `(-2, 3), (-2, 3), (-2, 3)`
- Toroidal priority: `false`

Group and Unit Cell

- Group: PG No. 18 D_3 32 (321 setting) [trigonal]
- Unit cell:
  $a =$ `1.00000`, $b =$ `1.00000`, $c =$ `1.00000`, $\alpha =$ `90.0`, $\beta =$ `90.0`, $\gamma =$ `120.0`
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ `[ 1.00000, 0.00000, 0.00000]`
  $\boldsymbol{a}_2 =$ `[-0.50000, 0.86603, 0.00000]`
  $\boldsymbol{a}_3 =$ `[ 0.00000, 0.00000, 1.00000]`

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	1	2	3^{+}_{001}	3	3^{-}_{001}	4	2_{110}	5	2_{100}
6	2_{010}								

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{G}_0(A_1)$	A_1	0	G, M	-	-	1
2	$\mathbb{Q}_0(A_1)$	A_1	0	Q, T	-	-	1
3	$\mathbb{G}_2(A_1)$	A_1	2	G, M	-	-	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
4	$\mathbb{Q}_2(A_1)$	A_1	2	Q, T	-	-	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
5	$\mathbb{G}_3(A_1)$	A_1	3	G, M	-	-	$\frac{\sqrt{10}x\left(x^2-3y^2\right)}{4}$
6	$\mathbb{Q}_3(A_1)$	A_1	3	Q, T	-	-	$\frac{\sqrt{10}x\left(x^2-3y^2\right)}{4}$
7	$\mathbb{G}_4(A_1,1)$	A_1	4	G, M	1	-	$\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$
8	$\mathbb{G}_4(A_1,2)$	A_1	4	G, M	2	-	$\frac{\sqrt{70}yz\left(3x^2-y^2\right)}{4}$
9	$\mathbb{Q}_4(A_1,2)$	A_1	4	Q, T	2	-	$\frac{\sqrt{70}yz\left(3x^2-y^2\right)}{4}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
10	$\mathbb{G}_1(A_2)$	A_2	1	G, M	-	-	z
11	$\mathbb{Q}_1(A_2)$	A_2	1	Q, T	-	-	z
12	$\mathbb{G}_3(A_2,1)$	A_2	3	G, M	1	-	$-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$
13	$\mathbb{G}_3(A_2,2)$	A_2	3	G, M	2	-	$\frac{\sqrt{10}y\left(3x^2-y^2\right)}{4}$
14	$\mathbb{Q}_3(A_2,1)$	A_2	3	Q, T	1	-	$-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$
15	$\mathbb{Q}_3(A_2,2)$	A_2	3	Q, T	2	-	$\frac{\sqrt{10}y\left(3x^2-y^2\right)}{4}$
16	$\mathbb{G}_4(A_2)$	A_2	4	G, M	-	-	$\frac{\sqrt{70}xz\left(x^2-3y^2\right)}{4}$
17	$\mathbb{Q}_4(A_2)$	A_2	4	Q, T	-	-	$\frac{\sqrt{70}xz\left(x^2-3y^2\right)}{4}$
18	$\mathbb{G}_{1,1}(E)$	E	1	G, M	-	1	x
19	$\mathbb{G}_{1,2}(E)$					2	y
20	$\mathbb{Q}_{1,1}(E)$	E	1	Q, T	-	1	x
21	$\mathbb{Q}_{1,2}(E)$					2	y
22	$\mathbb{G}_{2,1}(E,1)$	E	2	G, M	1	1	$\sqrt{3}yz$
23	$\mathbb{G}_{2,2}(E,1)$					2	$-\sqrt{3}xz$
24	$\mathbb{G}_{2,1}(E,2)$	E	2	G, M	2	1	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
25	$\mathbb{G}_{2,2}(E,2)$					2	$-\sqrt{3}xy$
26	$\mathbb{Q}_{2,1}(E,1)$	E	2	Q, T	1	1	$\sqrt{3}yz$
27	$\mathbb{Q}_{2,2}(E,1)$					2	$-\sqrt{3}xz$
28	$\mathbb{Q}_{2,1}(E,2)$	E	2	Q, T	2	1	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
29	$\mathbb{Q}_{2,2}(E,2)$					2	$-\sqrt{3}xy$
30	$\mathbb{G}_{3,1}(E,1)$	E	3	G, M	1	1	$-\frac{\sqrt{6}x\left(x^2+y^2-4z^2\right)}{4}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
31	$\mathbb{G}_{3,2}(E,1)$					2	$-\frac{\sqrt{6}y\left(x^2+y^2-4z^2\right)}{4}$
32	$\mathbb{G}_{3,1}(E,2)$	E	3	G, M	2	1	$\sqrt{15}xyz$
33	$\mathbb{G}_{3,2}(E,2)$					2	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
34	$\mathbb{Q}_{3,1}(E,1)$	E	3	Q, T	1	1	$-\frac{\sqrt{6}x\left(x^2+y^2-4z^2\right)}{4}$
35	$\mathbb{Q}_{3,2}(E,1)$					2	$-\frac{\sqrt{6}y\left(x^2+y^2-4z^2\right)}{4}$
36	$\mathbb{Q}_{3,1}(E,2)$	E	3	Q, T	2	1	$\sqrt{15}xyz$
37	$\mathbb{Q}_{3,2}(E,2)$					2	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
38	$\mathbb{G}_{4,1}(E,1)$	E	4	G, M	1	1	$-\frac{\sqrt{10}yz\left(3x^2+3y^2-4z^2\right)}{4}$
39	$\mathbb{G}_{4,2}(E,1)$					2	$\frac{\sqrt{10}xz\left(3x^2+3y^2-4z^2\right)}{4}$
40	$\mathbb{G}_{4,1}(E,2)$	E	4	G, M	2	1	$\frac{\sqrt{35}\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{8}$
41	$\mathbb{G}_{4,2}(E,2)$					2	$\frac{\sqrt{35}xy(x-y)(x+y)}{2}$
42	$\mathbb{G}_{4,1}(E,3)$	E	4	G, M	3	1	$-\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$
43	$\mathbb{G}_{4,2}(E,3)$					2	$\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$
44	$\mathbb{Q}_{4,1}(E,1)$	E	4	Q, T	1	1	$-\frac{\sqrt{10}yz\left(3x^2+3y^2-4z^2\right)}{4}$
45	$\mathbb{Q}_{4,2}(E,1)$					2	$\frac{\sqrt{10}xz\left(3x^2+3y^2-4z^2\right)}{4}$
46	$\mathbb{Q}_{4,1}(E,2)$	E	4	Q, T	2	1	$\frac{\sqrt{35}\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{8}$
47	$\mathbb{Q}_{4,2}(E,2)$					2	$\frac{\sqrt{35}xy(x-y)(x+y)}{2}$
48	$\mathbb{Q}_{4,1}(E,3)$	E	4	Q, T	3	1	$-\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$
49	$\mathbb{Q}_{4,2}(E,3)$					2	$\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$

Basis in full matrix

Table 3: dimension = 48

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$\|s,\uparrow\rangle$ @A(1)	1	$\|s,\downarrow\rangle$ @A(1)	2	$\|p_x,\uparrow\rangle$ @A(1)	3	$\|p_x,\downarrow\rangle$ @A(1)	4	$\|p_y,\uparrow\rangle$ @A(1)
5	$\|p_y,\downarrow\rangle$ @A(1)	6	$\|p_z,\uparrow\rangle$ @A(1)	7	$\|p_z,\downarrow\rangle$ @A(1)	8	$\|s,\uparrow\rangle$ @A(2)	9	$\|s,\downarrow\rangle$ @A(2)
10	$\|p_x,\uparrow\rangle$ @A(2)	11	$\|p_x,\downarrow\rangle$ @A(2)	12	$\|p_y,\uparrow\rangle$ @A(2)	13	$\|p_y,\downarrow\rangle$ @A(2)	14	$\|p_z,\uparrow\rangle$ @A(2)
15	$\|p_z,\downarrow\rangle$ @A(2)	16	$\|s,\uparrow\rangle$ @A(3)	17	$\|s,\downarrow\rangle$ @A(3)	18	$\|p_x,\uparrow\rangle$ @A(3)	19	$\|p_x,\downarrow\rangle$ @A(3)
20	$\|p_y,\uparrow\rangle$ @A(3)	21	$\|p_y,\downarrow\rangle$ @A(3)	22	$\|p_z,\uparrow\rangle$ @A(3)	23	$\|p_z,\downarrow\rangle$ @A(3)	24	$\|s,\uparrow\rangle$ @A(4)
25	$\|s,\downarrow\rangle$ @A(4)	26	$\|p_x,\uparrow\rangle$ @A(4)	27	$\|p_x,\downarrow\rangle$ @A(4)	28	$\|p_y,\uparrow\rangle$ @A(4)	29	$\|p_y,\downarrow\rangle$ @A(4)
30	$\|p_z,\uparrow\rangle$ @A(4)	31	$\|p_z,\downarrow\rangle$ @A(4)	32	$\|s,\uparrow\rangle$ @A(5)	33	$\|s,\downarrow\rangle$ @A(5)	34	$\|p_x,\uparrow\rangle$ @A(5)
35	$\|p_x,\downarrow\rangle$ @A(5)	36	$\|p_y,\uparrow\rangle$ @A(5)	37	$\|p_y,\downarrow\rangle$ @A(5)	38	$\|p_z,\uparrow\rangle$ @A(5)	39	$\|p_z,\downarrow\rangle$ @A(5)
40	$\|s,\uparrow\rangle$ @A(6)	41	$\|s,\downarrow\rangle$ @A(6)	42	$\|p_x,\uparrow\rangle$ @A(6)	43	$\|p_x,\downarrow\rangle$ @A(6)	44	$\|p_y,\uparrow\rangle$ @A(6)
45	$\|p_y,\downarrow\rangle$ @A(6)	46	$\|p_z,\uparrow\rangle$ @A(6)	47	$\|p_z,\downarrow\rangle$ @A(6)				

Table 4: Atomic basis (orbital part only)

orbital	definition
$\|s\rangle$	1
$\|p_x\rangle$	x
$\|p_y\rangle$	y
$\|p_z\rangle$	z

SAMB: 744 (all 744)

- `A` : '`A`' site-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `6c`

$\boxed{\text{z1}}$ $\mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

$\boxed{\text{z129}}$ $\mathbb{Q}_1^{(c)}(A_2) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_1^{(s)}(A_2)$

$\boxed{\text{z249}}$ $\mathbb{Q}_{1,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z250}}$ $\mathbb{Q}_{1,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$

$\boxed{\text{z251}}$ $\mathbb{Q}_{2,1}^{(c)}(E,1) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z252}}$ $\mathbb{Q}_{2,2}^{(c)}(E,1) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$

- `A` : '`A`' site-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `6c`

$\boxed{\text{z2}}$ $\mathbb{Q}_0^{(c)}(A_1) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{3} + \frac{\sqrt{3}\mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_1^{(s)}(A_2)}{3}$

$\boxed{\text{z3}}$ $\mathbb{Q}_2^{(c)}(A_1) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\sqrt{6}\mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_1^{(s)}(A_2)}{3}$

$\boxed{\text{z4}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_1) = -\frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$

$\boxed{\text{z5}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_1) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$

$\boxed{\text{z6}}$ $\mathbb{Q}_0^{(1,0;c)}(A_1) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1}^{(s)}(A_2)}{3}$

$\boxed{\text{z7}}$ $\mathbb{Q}_2^{(1,0;c)}(A_1) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1}^{(s)}(A_2)}{3}$

$\boxed{\text{z8}}$ $\mathbb{G}_2^{(c)}(A_1) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$

$\boxed{\text{z9}}$ $\mathbb{G}_0^{(1,-1;c)}(A_1) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$

$\boxed{\text{z10}}$ $\mathbb{G}_2^{(1,-1;c)}(A_1) = \mathbb{G}_2^{(1,-1;a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

$\boxed{\text{z11}}$ $\mathbb{G}_3^{(1,-1;c)}(A_1) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{2} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$

$\boxed{\text{z12}}$ $\mathbb{G}_2^{(1,0;c)}(A_1) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$

$\boxed{\text{z13}}$ $\mathbb{G}_0^{(1,1;c)}(A_1) = \mathbb{G}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$

$\boxed{\text{z130}}$ $\mathbb{Q}_1^{(c)}(A_2,a) = \mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_0^{(s)}(A_1)$

$\boxed{\text{z131}}$ $\mathbb{Q}_1^{(c)}(A_2,b) = -\frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z132}}$ $\mathbb{Q}_1^{(1,-1;c)}(A_2) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z133}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_2,2) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z134}}$ $\mathbb{Q}_1^{(1,0;c)}(A_2,a) = \mathbb{Q}_1^{(1,0;a)}(A_2)\mathbb{Q}_0^{(s)}(A_1)$

$\boxed{\text{z135}}$ $\mathbb{Q}_1^{(1,0;c)}(A_2,b) = -\frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z136}}$ $\mathbb{G}_1^{(c)}(A_2) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z137}}$ $\mathbb{G}_1^{(1,-1;c)}(A_2) = \frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{10} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{10} + \frac{\sqrt{10}\mathbb{G}_2^{(1,-1;a)}(A_1)\mathbb{Q}_1^{(s)}(A_2)}{5}$

$\boxed{\text{z138}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2,1) = -\frac{\sqrt{5}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{5} + \frac{\sqrt{15}\mathbb{G}_2^{(1,-1;a)}(A_1)\mathbb{Q}_1^{(s)}(A_2)}{5}$

$\boxed{\text{z139}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2,2) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z140}}$ $\mathbb{G}_1^{(1,0;c)}(A_2) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z141}}$ $\mathbb{G}_1^{(1,1;c)}(A_2) = \mathbb{G}_0^{(1,1;a)}(A_1)\mathbb{Q}_1^{(s)}(A_2)$

$\boxed{\text{z253}}$ $\mathbb{Q}_{1,1}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_0^{(s)}(A_1)}{2}$

$\boxed{\text{z254}}$ $\mathbb{Q}_{1,2}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_0^{(s)}(A_1)}{2}$

$\boxed{\text{z255}}$ $\mathbb{Q}_{1,1}^{(c)}(E,b) = -\frac{\sqrt{2}\mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$

$\boxed{\text{z256}}$ $\mathbb{Q}_{1,2}^{(c)}(E,b) = \frac{\sqrt{2}\mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z257}}$ $\mathbb{Q}_{2,1}^{(c)}(E,1) = \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_1^{(s)}(A_2)}{2} + \frac{\mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$

$\boxed{\text{z258}}$ $\mathbb{Q}_{2,2}^{(c)}(E,1) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_1^{(s)}(A_2)}{2} - \frac{\mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z259}}$ $\mathbb{Q}_{2,1}^{(c)}(E,2) = \frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$

$\boxed{\text{z260}}$ $\mathbb{Q}_{2,2}^{(c)}(E,2) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z261}}$ $\mathbb{Q}_{3,1}^{(c)}(E,2) = \frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$

$\boxed{\text{z262}}$ $$\mathbb{Q}_{3,2}^{(c)}(E,2) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$$

$\boxed{\text{z263}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(E) = -\frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{10} + \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{10} - \frac{\sqrt{30}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{10}$$

$\boxed{\text{z264}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(E) = \frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{10} + \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{10} - \frac{\sqrt{30}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{10}$$

$\boxed{\text{z265}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{6} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$\boxed{\text{z266}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{6} + \frac{\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z267}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{3}$$

$\boxed{\text{z268}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6}$$

$\boxed{\text{z269}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,1) = \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{10} - \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{5}$$

$\boxed{\text{z270}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,1) = -\frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{10} - \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{5}$$

$\boxed{\text{z271}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z272}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z273}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(E,b) = -\frac{\sqrt{2}\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z274}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(E,b) = \frac{\sqrt{2}\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$$

$\boxed{\text{z275}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,1) = \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(s)}(A_2)}{2} + \frac{\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z276}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,1) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(s)}(A_2)}{2} - \frac{\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$\boxed{\text{z277}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,2) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z278}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,2) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$\boxed{\text{z279}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E,2) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z280}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E,2) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$$

$\boxed{\text{z281}}$ $$\mathbb{G}_{1,1}^{(c)}(E) = \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1}^{(s)}(A_2)}{2} - \frac{\mathbb{Q}_{1}^{(a)}(A_2)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z282}}$ $$\mathbb{G}_{1,2}^{(c)}(E) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1}^{(s)}(A_2)}{2} + \frac{\mathbb{Q}_{1}^{(a)}(A_2)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$\boxed{\text{z283}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(E) = \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{10} - \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{10}$$

$\boxed{\text{z284}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(E) = \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{10} - \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{10}$$

$\boxed{\text{z285}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,1) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z286}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,1) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z287}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,2a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z288}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,2a) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z289}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,2b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z290}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E,2b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} + \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z291}}$ $\mathbb{G}_{3,1}^{(1,-1;c)}(E,1) = -\frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{30} - \frac{2\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{30} + \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{5}$

$\boxed{\text{z292}}$ $\mathbb{G}_{3,2}^{(1,-1;c)}(E,1) = \frac{2\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{30} + \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{30} + \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{5}$

$\boxed{\text{z293}}$ $\mathbb{G}_{3,1}^{(1,-1;c)}(E,2) = \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6} - \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} - \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{6}$

$\boxed{\text{z294}}$ $\mathbb{G}_{3,2}^{(1,-1;c)}(E,2) = -\frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{6} - \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6}$

$\boxed{\text{z295}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(E) = \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(s)}(A_2)}{2} - \frac{\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$

$\boxed{\text{z296}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(E) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(s)}(A_2)}{2} + \frac{\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z297}}$ $\mathbb{G}_{1,1}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z298}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$

$\boxed{\text{z299}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(E,1) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z300}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E,1) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$

- `A` : '`A`' site-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `6c`

  $\boxed{\text{z14}}$ $\mathbb{Q}_{0}^{(c)}(A_1,a) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(s)}(A_1)$

$\boxed{\text{z15}}$ $$\mathbb{Q}_0^{(c)}(A_1,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z16}}$ $$\mathbb{Q}_2^{(c)}(A_1) = \mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$$

$\boxed{\text{z17}}$ $$\mathbb{Q}_3^{(c)}(A_1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z18}}$ $$\mathbb{Q}_4^{(c)}(A_1,2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z19}}$ $$\mathbb{Q}_0^{(1,-1;c)}(A_1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z20}}$ $$\mathbb{Q}_2^{(1,-1;c)}(A_1) = \mathbb{Q}_2^{(1,-1;a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$$

$\boxed{\text{z21}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z22}}$ $$\mathbb{Q}_4^{(1,-1;c)}(A_1,2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z23}}$ $$\mathbb{Q}_2^{(1,0;c)}(A_1) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z24}}$ $$\mathbb{Q}_0^{(1,1;c)}(A_1) = \mathbb{Q}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$$

$\boxed{\text{z25}}$ $$\mathbb{G}_2^{(c)}(A_1) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z26}}$ $$\mathbb{G}_2^{(1,-1;c)}(A_1) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z27}}$ $$\mathbb{G}_0^{(1,0;c)}(A_1) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{3} + \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{3} + \frac{\sqrt{3}\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1}^{(s)}(A_2)}{3}$$

$\boxed{\text{z28}}$ $$\mathbb{G}_2^{(1,0;c)}(A_1) = -\frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{6} - \frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\sqrt{6}\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1}^{(s)}(A_2)}{3}$$

$\boxed{\text{z142}}$ $\mathbb{Q}_1^{(c)}(A_2,a) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_1^{(s)}(A_2)$

$\boxed{\text{z143}}$ $\mathbb{Q}_1^{(c)}(A_2,b) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{10} + \frac{\sqrt{10}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_1^{(s)}(A_2)}{5}$

$\boxed{\text{z144}}$ $\mathbb{Q}_3^{(c)}(A_2,1) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{5} + \frac{\sqrt{15}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_1^{(s)}(A_2)}{5}$

$\boxed{\text{z145}}$ $\mathbb{Q}_3^{(c)}(A_2,2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z146}}$ $\mathbb{Q}_4^{(c)}(A_2) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z147}}$ $\mathbb{Q}_1^{(1,-1;c)}(A_2) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{10} + \frac{\sqrt{10}\mathbb{Q}_2^{(1,-1;a)}(A_1)\mathbb{Q}_1^{(s)}(A_2)}{5}$

$\boxed{\text{z148}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_2,1) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{5} + \frac{\sqrt{15}\mathbb{Q}_2^{(1,-1;a)}(A_1)\mathbb{Q}_1^{(s)}(A_2)}{5}$

$\boxed{\text{z149}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_2,2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z150}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_2) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z151}}$ $\mathbb{Q}_1^{(1,0;c)}(A_2) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z152}}$ $\mathbb{Q}_1^{(1,1;c)}(A_2) = \mathbb{Q}_0^{(1,1;a)}(A_1)\mathbb{Q}_1^{(s)}(A_2)$

$\boxed{\text{z153}}$ $\mathbb{G}_1^{(c)}(A_2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z154}}$ $\mathbb{G}_1^{(1,-1;c)}(A_2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$

$\boxed{\text{z155}}$ $\mathbb{G}_1^{(1,0;c)}(A_2,a) = \mathbb{G}_1^{(1,0;a)}(A_2)\mathbb{Q}_0^{(s)}(A_1)$

$\boxed{\text{z156}}$ $$\mathbb{G}_1^{(1,0;c)}(A_2,b) = -\frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$$

$\boxed{\text{z301}}$ $$\mathbb{Q}_{1,1}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$\boxed{\text{z302}}$ $$\mathbb{Q}_{1,2}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z303}}$ $$\mathbb{Q}_{1,1}^{(c)}(E,b) = \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_1^{(s)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{10} - \frac{\sqrt{5}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{10}$$

$\boxed{\text{z304}}$ $$\mathbb{Q}_{1,2}^{(c)}(E,b) = \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_1^{(s)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{10} - \frac{\sqrt{5}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{10}$$

$\boxed{\text{z305}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$$

$\boxed{\text{z306}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z307}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,1b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$\boxed{\text{z308}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,1b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$\boxed{\text{z309}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,1c) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{14} + \frac{\sqrt{14}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{14}$$

$\boxed{\text{z310}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,1c) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{14} + \frac{\sqrt{14}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{14}$$

$\boxed{\text{z311}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$\boxed{\text{z312}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$\boxed{\text{z313}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,2b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z314}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,2b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$$

$\boxed{\text{z315}}$ $$\mathbb{Q}_{3,1}^{(c)}(E,1) = -\frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{30} - \frac{2\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{30} + \frac{\sqrt{5}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{5}$$

$\boxed{\text{z316}}$ $$\mathbb{Q}_{3,2}^{(c)}(E,1) = \frac{2\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{30} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{30} + \frac{\sqrt{5}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{5}$$

$\boxed{\text{z317}}$ $$\mathbb{Q}_{3,1}^{(c)}(E,2) = \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{6}$$

$\boxed{\text{z318}}$ $$\mathbb{Q}_{3,2}^{(c)}(E,2) = -\frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6}$$

$\boxed{\text{z319}}$ $$\mathbb{Q}_{4,1}^{(c)}(E,1) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{7}$$

$\boxed{\text{z320}}$ $$\mathbb{Q}_{4,2}^{(c)}(E,1) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{7}$$

$\boxed{\text{z321}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(E) = \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{10} - \frac{\sqrt{5}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{10}$$

$\boxed{\text{z322}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(E) = \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{10} - \frac{\sqrt{5}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{10}$$

$\boxed{\text{z323}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z324}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z325}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{14}$$

$\boxed{\text{z326}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1b) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{14}$$

$\boxed{\text{z327}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z328}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z329}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2b) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z330}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2b) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$$

$\boxed{\text{z331}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,1) = -\frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{30} - \frac{2\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{30} + \frac{\sqrt{5}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{5}$$

$\boxed{\text{z332}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,1) = \frac{2\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{30} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{30} + \frac{\sqrt{5}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{5}$$

$\boxed{\text{z333}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,2) = \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{6}$$

$\boxed{\text{z334}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,2) = -\frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6}$$

$\boxed{\text{z335}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E,1) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{7}$$

$\boxed{\text{z336}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E,1) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{7}$$

$\boxed{\text{z337}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(E) = \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(s)}(A_2)}{2} - \frac{\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z338}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(E) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(s)}(A_2)}{2} + \frac{\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$\boxed{\text{z339}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,1) = -\frac{\sqrt{2}\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z340}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,1) = \frac{\sqrt{2}\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$$

$\boxed{\text{z341}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,2) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z342}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,2) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$$

$\boxed{\text{z343}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$\boxed{\text{z344}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z345}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(E,1) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,1}^{(s)}(E,1)}{2}$$

$\boxed{\text{z346}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(E,1) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,2}^{(s)}(E,1)}{2}$$

$\boxed{\text{z347}}$ $$\mathbb{G}_{2,1}^{(c)}(E,1) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$\boxed{\text{z348}}$ $$\mathbb{G}_{2,2}^{(c)}(E,1) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{6} + \frac{\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z349}}$ $$\mathbb{G}_{2,1}^{(c)}(E,2) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{3}$$

$\boxed{\text{z350}}$ $$\mathbb{G}_{2,2}^{(c)}(E,2) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6}$$

$\boxed{\text{z351}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,1) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$\boxed{\text{z352}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,1) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(s)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(s)}(E)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(s)}(E)}{6} + \frac{\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$\boxed{\text{z353}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,2) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{3}$$

$\boxed{\text{z354}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,2) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(s)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(s)}(A_2)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(s)}(E)}{6}$$

$\boxed{\text{z355}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z356}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$

$\boxed{\text{z357}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(E,1) = \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(s)}(A_2)}{2} + \frac{\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$

$\boxed{\text{z358}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(E,1) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(s)}(A_2)}{2} - \frac{\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

$\boxed{\text{z359}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(E,2) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$

$\boxed{\text{z360}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(E,2) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$

- `A;A_001_1` : 'A'-'A' bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `6a@6c`

$\boxed{\text{z29}}$ $\mathbb{Q}_{0}^{(c)}(A_1) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z30}}$ $\mathbb{Q}_{0}^{(1,-1;c)}(A_1) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z31}}$ $\mathbb{G}_{0}^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z32}}$ $\mathbb{G}_{2}^{(1,-1;c)}(A_1) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z157}}$ $\mathbb{Q}_{1}^{(c)}(A_2) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{1}^{(b)}(A_2)$

$\boxed{\text{z158}}$ $\mathbb{Q}_{1}^{(1,-1;c)}(A_2) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z159}}$ $\mathbb{G}_{1}^{(1,-1;c)}(A_2,a) = \mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{0}^{(b)}(A_1)$

$\boxed{\text{z160}}$ $\mathbb{G}_{1}^{(1,-1;c)}(A_2,b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z361}}$ $\mathbb{Q}_{1,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z362}}$ $\mathbb{Q}_{1,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z363}}$ $\mathbb{Q}_{2,1}^{(c)}(E,1) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z364}}$ $\mathbb{Q}_{2,2}^{(c)}(E,1) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z365}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(E) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z366}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(E) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z367}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1) = \frac{\sqrt{2}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z368}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z369}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z370}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z371}}$ $\mathbb{G}_{1,1}^{(1,-1;c)}(E) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z372}}$ $\mathbb{G}_{1,2}^{(1,-1;c)}(E) = \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z373}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E,1) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z374}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E,1) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z375}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E,2) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z376}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E,2) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

- `A;A_001_1` : `'A'-'A'` bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `6a@6c`

$\boxed{\text{z33}}$ $\mathbb{Q}_{0}^{(c)}(A_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1}^{(a)}(A_2)\mathbb{Q}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z34}}$ $\mathbb{Q}_{0}^{(c)}(A_1,b) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z35}}$ $\mathbb{Q}_{2}^{(c)}(A_1,a) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1}^{(a)}(A_2)\mathbb{Q}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z36}}$ $\mathbb{Q}_{2}^{(c)}(A_1,b) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z37}}$ $\mathbb{Q}_{2}^{(1,-1;c)}(A_1,a) = -\frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z38}}$ $\mathbb{Q}_{2}^{(1,-1;c)}(A_1,b) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z39}}$ $\mathbb{Q}_{3}^{(1,-1;c)}(A_1,a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z40}}$ $\mathbb{Q}_{3}^{(1,-1;c)}(A_1,b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z41}}$ $\mathbb{Q}_{0}^{(1,0;c)}(A_1,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z42}}$ $\mathbb{Q}_{0}^{(1,0;c)}(A_1,b) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z43}}$ $\mathbb{Q}_2^{(1,0;c)}(A_1,a) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z44}}$ $\mathbb{Q}_2^{(1,0;c)}(A_1,b) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z45}}$ $\mathbb{G}_0^{(c)}(A_1) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{T}_{1,2}^{(a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z46}}$ $\mathbb{G}_2^{(c)}(A_1) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z47}}$ $\mathbb{G}_0^{(1,-1;c)}(A_1) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z48}}$ $\mathbb{G}_2^{(1,-1;c)}(A_1,a) = \mathbb{G}_2^{(1,-1;a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z49}}$ $\mathbb{G}_2^{(1,-1;c)}(A_1,b) = \mathbb{M}_2^{(1,-1;a)}(A_1)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z50}}$ $\mathbb{G}_2^{(1,-1;c)}(A_1,c) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z51}}$ $\mathbb{G}_3^{(1,-1;c)}(A_1,a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z52}}$ $\mathbb{G}_3^{(1,-1;c)}(A_1,b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z53}}$ $\mathbb{G}_0^{(1,0;c)}(A_1) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z54}}$ $\mathbb{G}_2^{(1,0;c)}(A_1) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z55}}$ $\mathbb{G}_0^{(1,1;c)}(A_1,a) = \mathbb{G}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z56}}$ $\mathbb{G}_0^{(1,1;c)}(A_1,b) = \mathbb{M}_0^{(1,1;a)}(A_1)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z161}}$ $\mathbb{Q}_1^{(c)}(A_2,a) = \mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z162}}$ $\mathbb{Q}_1^{(c)}(A_2,b) = -\frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z163}}$ $\mathbb{Q}_1^{(c)}(A_2,c) = \mathbb{T}_1^{(a)}(A_2)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z164}}$ $\mathbb{Q}_1^{(c)}(A_2,d) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z165}}$ $\mathbb{Q}_1^{(1,-1;c)}(A_2,a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z166}}$ $\mathbb{Q}_1^{(1,-1;c)}(A_2,b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z167}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_2,2a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z168}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_2,2b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z169}}$ $\mathbb{Q}_1^{(1,0;c)}(A_2,a) = \mathbb{Q}_1^{(1,0;a)}(A_2)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z170}}$ $\mathbb{Q}_1^{(1,0;c)}(A_2,b) = -\frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z171}}$ $\mathbb{Q}_1^{(1,0;c)}(A_2,c) = \mathbb{T}_1^{(1,0;a)}(A_2)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z172}}$ $\mathbb{Q}_1^{(1,0;c)}(A_2,d) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z173}}$ $\mathbb{G}_1^{(c)}(A_2,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z174}}$ $\mathbb{G}_1^{(c)}(A_2,b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z175}}$ $\mathbb{G}_1^{(1,-1;c)}(A_2,a) = \frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{10} + \frac{\sqrt{10}\mathbb{G}_2^{(1,-1;a)}(A_1)\mathbb{Q}_1^{(b)}(A_2)}{5}$

$\boxed{\text{z176}}$ $\mathbb{G}_1^{(1,-1;c)}(A_2,b) = \frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{10} + \frac{\sqrt{10}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1}^{(b)}(A_2)}{5}$

$\boxed{\text{z177}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2,1a) = -\frac{\sqrt{5}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{5} + \frac{\sqrt{15}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1}^{(b)}(A_2)}{5}$

$\boxed{\text{z178}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2,1b) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{5} + \frac{\sqrt{15}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1}^{(b)}(A_2)}{5}$

$\boxed{\text{z179}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2,2a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z180}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2,2b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z181}}$ $\mathbb{G}_1^{(1,0;c)}(A_2,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z182}}$ $\mathbb{G}_1^{(1,0;c)}(A_2,b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z183}}$ $\mathbb{G}_1^{(1,1;c)}(A_2,a) = \mathbb{G}_0^{(1,1;a)}(A_1)\mathbb{Q}_1^{(b)}(A_2)$

$\boxed{\text{z184}}$ $\mathbb{G}_1^{(1,1;c)}(A_2,b) = \mathbb{M}_0^{(1,1;a)}(A_1)\mathbb{T}_1^{(b)}(A_2)$

$\boxed{\text{z377}}$ $\mathbb{Q}_{1,1}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_0^{(b)}(A_1)}{2}$

$\boxed{\text{z378}}$ $\mathbb{Q}_{1,2}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_0^{(b)}(A_1)}{2}$

$\boxed{\text{z379}}$ $\mathbb{Q}_{1,1}^{(c)}(E,b) = -\frac{\sqrt{2}\mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z380}}$ $\mathbb{Q}_{1,2}^{(c)}(E,b) = \frac{\sqrt{2}\mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z381}}$ $\mathbb{Q}_{1,1}^{(c)}(E,c) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_0^{(b)}(A_1)}{2}$

$\boxed{\text{z382}}$ $\mathbb{Q}_{1,2}^{(c)}(E,c) = \frac{\sqrt{2}\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z383}}$ $\mathbb{Q}_{1,1}^{(c)}(E,d) = -\frac{\sqrt{2}\mathbb{T}_{1}^{(a)}(A_2)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z384}}$ $\mathbb{Q}_{1,2}^{(c)}(E,d) = \frac{\sqrt{2}\mathbb{T}_{1}^{(a)}(A_2)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z385}}$ $\mathbb{Q}_{2,1}^{(c)}(E,1a) = \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{Q}_{1}^{(a)}(A_2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z386}}$ $\mathbb{Q}_{2,2}^{(c)}(E,1a) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{Q}_{1}^{(a)}(A_2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z387}}$ $\mathbb{Q}_{2,1}^{(c)}(E,1b) = \frac{\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z388}}$ $\mathbb{Q}_{2,2}^{(c)}(E,1b) = -\frac{\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z389}}$ $\mathbb{Q}_{2,1}^{(c)}(E,2a) = \frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z390}}$ $\mathbb{Q}_{2,2}^{(c)}(E,2a) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z391}}$ $\mathbb{Q}_{2,1}^{(c)}(E,2b) = \frac{\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z392}}$ $\mathbb{Q}_{2,2}^{(c)}(E,2b) = -\frac{\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z393}}$ $\mathbb{Q}_{3,1}^{(c)}(E,2) = \frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z394}}$ $\mathbb{Q}_{3,2}^{(c)}(E,2) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z395}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(E,a) = -\frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{10} + \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{10} - \frac{\sqrt{30}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{10}$

$\boxed{\text{z396}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(E,a) = \frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{10} + \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{10} - \frac{\sqrt{30}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{10}$$

$\boxed{\text{z397}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(E,b) = \frac{\sqrt{42}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{42}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{14}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{M}_{1,1}^{(b)}(E)}{14}$$

$\boxed{\text{z398}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(E,b) = -\frac{\sqrt{42}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{42}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{14}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{M}_{1,2}^{(b)}(E)}{14}$$

$\boxed{\text{z399}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1a) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z400}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1a) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{6} + \frac{\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z401}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1b) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z402}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z403}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2a) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z404}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2a) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z405}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2b) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z406}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2b) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z407}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,1a) = \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{10} - \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{5}$$

$\boxed{\text{z408}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,1a) = -\frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{10} - \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{5}$$

$\boxed{\text{z409}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,1b) = \frac{\sqrt{7}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{7}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{14} + \frac{\sqrt{21}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{M}_{1,1}^{(b)}(E)}{7}$$

$\boxed{\text{z410}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{7}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{7}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{14} + \frac{\sqrt{21}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{M}_{1,2}^{(b)}(E)}{7}$

$\boxed{\text{z411}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(E,2) = \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z412}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E,2) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z413}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z414}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z415}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(E,b) = -\frac{\sqrt{2}\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z416}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(E,b) = \frac{\sqrt{2}\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z417}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(E,c) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z418}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(E,c) = \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z419}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(E,d) = -\frac{\sqrt{2}\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z420}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(E,d) = \frac{\sqrt{2}\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z421}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E,1a) = \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z422}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E,1a) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z423}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E,1b) = \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z424}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,1b) = -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z425}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,2a) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z426}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,2a) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z427}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,2b) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z428}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,2b) = -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z429}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E,2) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$$

$\boxed{\text{z430}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E,2) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$$

$\boxed{\text{z431}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{M}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z432}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{M}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z433}}$ $$\mathbb{G}_{1,1}^{(c)}(E,a) = \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{Q}_{1}^{(a)}(A_2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z434}}$ $$\mathbb{G}_{1,2}^{(c)}(E,a) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{Q}_{1}^{(a)}(A_2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z435}}$ $$\mathbb{G}_{1,1}^{(c)}(E,b) = \frac{\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z436}}$ $$\mathbb{G}_{1,2}^{(c)}(E,b) = -\frac{\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z437}}$ $$\mathbb{G}_{2,1}^{(c)}(E,2) = \frac{\mathbb{T}_{1,1}^{(a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z438}}$ $$\mathbb{G}_{2,2}^{(c)}(E,2) = -\frac{\mathbb{T}_{1,1}^{(a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z439}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(E,a) = \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{10}$$

$\boxed{\text{z440}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(E,a) = \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{10}$$

$\boxed{\text{z441}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(E,b) = \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{10}$$

$\boxed{\text{z442}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(E,b) = \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{10}$$

$\boxed{\text{z443}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,1a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z444}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,1a) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z445}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,1b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z446}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,1b) = \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z447}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,2a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z448}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,2a) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z449}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,2b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$$

$\boxed{\text{z450}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,2b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} + \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$$

$\boxed{\text{z451}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,2c) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z452}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,2c) = \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z453}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(E,1a) = -\frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{30} - \frac{2\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{5}$$

$\boxed{\text{z454}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(E,1a) = \frac{2\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{5}$$

$\boxed{\text{z455}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{30} - \frac{2\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{5}$$

$\boxed{\text{z456}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(E,1b) = \frac{2\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{5}$$

$\boxed{\text{z457}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(E,2a) = \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{6}$$

$\boxed{\text{z458}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(E,2a) = -\frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z459}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(E,2b) = \frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6}$$

$\boxed{\text{z460}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(E,2b) = -\frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{6}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z461}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(E,a) = \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z462}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(E,a) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z463}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(E,b) = \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z464}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(E,b) = -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z465}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E,2) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z466}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(E,2) = -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z467}}$ $\mathbb{G}_{1,1}^{(1,1;c)}(E,a) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z468}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(E,a) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z469}}$ $\mathbb{G}_{1,1}^{(1,1;c)}(E,b) = \frac{\sqrt{2}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z470}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(E,b) = \frac{\sqrt{2}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z471}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(E,1) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z472}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E,1) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

- `A;A_001_1` : `'A'-'A'` bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `6a@6c`

$\boxed{\text{z57}}$ $\mathbb{Q}_{0}^{(c)}(A_1,a) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z58}}$ $\mathbb{Q}_{0}^{(c)}(A_1,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z59}}$ $\mathbb{Q}_{0}^{(c)}(A_1,c) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z60}}$ $\mathbb{Q}_{2}^{(c)}(A_1) = \mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z61}}$ $\mathbb{Q}_{3}^{(c)}(A_1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z62}}$ $\mathbb{Q}_4^{(c)}(A_1,2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z63}}$ $\mathbb{Q}_0^{(1,-1;c)}(A_1,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z64}}$ $\mathbb{Q}_0^{(1,-1;c)}(A_1,b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z65}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_1,a) = \mathbb{Q}_2^{(1,-1;a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z66}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_1,b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z67}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_1,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z68}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_1,b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{4} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{4} + \frac{\sqrt{3}\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1}^{(b)}(A_2)}{2}$

$\boxed{\text{z69}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_1,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z70}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_1,2b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z71}}$ $\mathbb{Q}_2^{(1,0;c)}(A_1,a) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z72}}$ $\mathbb{Q}_2^{(1,0;c)}(A_1,b) = \mathbb{T}_2^{(1,0;a)}(A_1)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z73}}$ $\mathbb{Q}_2^{(1,0;c)}(A_1,c) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z74}}$ $\mathbb{Q}_3^{(1,0;c)}(A_1) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z75}}$ $\mathbb{Q}_0^{(1,1;c)}(A_1,a) = \mathbb{Q}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z76}}$ $\mathbb{Q}_0^{(1,1;c)}(A_1,b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z77}}$ $\mathbb{G}_0^{(c)}(A_1) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z78}}$ $\mathbb{G}_2^{(c)}(A_1,a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z79}}$ $\mathbb{G}_2^{(c)}(A_1,b) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z80}}$ $\mathbb{G}_0^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z81}}$ $\mathbb{G}_2^{(1,-1;c)}(A_1,a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z82}}$ $\mathbb{G}_2^{(1,-1;c)}(A_1,b) = \frac{\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{7} + \frac{\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{7} + \frac{\sqrt{21}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1}^{(b)}(A_2)}{7}$

$\boxed{\text{z83}}$ $\mathbb{G}_2^{(1,-1;c)}(A_1,c) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z84}}$ $\mathbb{G}_3^{(1,-1;c)}(A_1) = \mathbb{M}_3^{(1,-1;a)}(A_1)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z85}}$ $\mathbb{G}_4^{(1,-1;c)}(A_1,1) = -\frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{14} + \frac{2\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1}^{(b)}(A_2)}{7}$

$\boxed{\text{z86}}$ $\mathbb{G}_4^{(1,-1;c)}(A_1,2) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{4} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{4} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1}^{(b)}(A_2)}{2}$

$\boxed{\text{z87}}$ $\mathbb{G}_0^{(1,0;c)}(A_1) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z88}}$ $\mathbb{G}_2^{(1,0;c)}(A_1,a) = -\frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z89}}$ $\mathbb{G}_2^{(1,0;c)}(A_1,b) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z90}}$ $$\mathbb{G}_3^{(1,0;c)}(A_1) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z91}}$ $$\mathbb{G}_0^{(1,1;c)}(A_1) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z92}}$ $$\mathbb{G}_2^{(1,1;c)}(A_1) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z185}}$ $$\mathbb{Q}_1^{(c)}(A_2,a) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_1^{(b)}(A_2)$$

$\boxed{\text{z186}}$ $$\mathbb{Q}_1^{(c)}(A_2,b) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1}^{(b)}(A_2)}{5}$$

$\boxed{\text{z187}}$ $$\mathbb{Q}_1^{(c)}(A_2,c) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z188}}$ $$\mathbb{Q}_3^{(c)}(A_2,1) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{5} + \frac{\sqrt{15}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1}^{(b)}(A_2)}{5}$$

$\boxed{\text{z189}}$ $$\mathbb{Q}_3^{(c)}(A_2,2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z190}}$ $$\mathbb{Q}_4^{(c)}(A_2) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$$

$\boxed{\text{z191}}$ $$\mathbb{Q}_1^{(1,-1;c)}(A_2,a) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1}^{(b)}(A_2)}{5}$$

$\boxed{\text{z192}}$ $$\mathbb{Q}_1^{(1,-1;c)}(A_2,b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z193}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_2,1a) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{5} + \frac{\sqrt{15}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1}^{(b)}(A_2)}{5}$$

$\boxed{\text{z194}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_2,1b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z195}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_2,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z196}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_2,2b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{4} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{4} - \frac{\sqrt{3}\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1}^{(b)}(A_2)}{2}$

$\boxed{\text{z197}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_2,a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z198}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_2,b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z199}}$ $\mathbb{Q}_1^{(1,0;c)}(A_2,a) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z200}}$ $\mathbb{Q}_1^{(1,0;c)}(A_2,b) = \frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{10} + \frac{\sqrt{10}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1}^{(b)}(A_2)}{5}$

$\boxed{\text{z201}}$ $\mathbb{Q}_3^{(1,0;c)}(A_2,1) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{5} + \frac{\sqrt{15}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1}^{(b)}(A_2)}{5}$

$\boxed{\text{z202}}$ $\mathbb{Q}_3^{(1,0;c)}(A_2,2) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z203}}$ $\mathbb{Q}_1^{(1,1;c)}(A_2,a) = \mathbb{Q}_0^{(1,1;a)}(A_1)\mathbb{Q}_1^{(b)}(A_2)$

$\boxed{\text{z204}}$ $\mathbb{Q}_1^{(1,1;c)}(A_2,b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z205}}$ $\mathbb{G}_1^{(c)}(A_2,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z206}}$ $\mathbb{G}_1^{(c)}(A_2,b) = \mathbb{M}_1^{(a)}(A_2)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z207}}$ $\mathbb{G}_1^{(c)}(A_2,c) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z208}}$ $\mathbb{G}_1^{(1,-1;c)}(A_2,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z209}}$ $\mathbb{G}_1^{(1,-1;c)}(A_2,b) = \mathbb{M}_1^{(1,-1;a)}(A_2)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z210}}$ $\mathbb{G}_1^{(1,-1;c)}(A_2,c) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z211}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2,1a) = \mathbb{M}_3^{(1,-1;a)}(A_2,1)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z212}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2,1b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z213}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2,2) = \mathbb{M}_3^{(1,-1;a)}(A_2,2)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z214}}$ $\mathbb{G}_4^{(1,-1;c)}(A_2) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{4} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{4} + \frac{\mathbb{M}_3^{(1,-1;a)}(A_1)\mathbb{T}_1^{(b)}(A_2)}{2}$

$\boxed{\text{z215}}$ $\mathbb{G}_1^{(1,0;c)}(A_2,a) = \mathbb{G}_1^{(1,0;a)}(A_2)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z216}}$ $\mathbb{G}_1^{(1,0;c)}(A_2,b) = -\frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} + \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z217}}$ $\mathbb{G}_1^{(1,0;c)}(A_2,c) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z218}}$ $\mathbb{G}_3^{(1,0;c)}(A_2,2) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z219}}$ $\mathbb{G}_1^{(1,1;c)}(A_2,a) = \mathbb{M}_1^{(1,1;a)}(A_2)\mathbb{T}_0^{(b)}(A_1)$

$\boxed{\text{z220}}$ $\mathbb{G}_1^{(1,1;c)}(A_2,b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z473}}$ $\mathbb{Q}_{1,1}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z474}}$ $\mathbb{Q}_{1,2}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z475}}$ $\mathbb{Q}_{1,1}^{(c)}(E,b) = \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_1^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{10}$

$\boxed{\text{z476}}$ $\mathbb{Q}_{1,2}^{(c)}(E,b) = \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_1^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{10}$

$\boxed{\text{z477}}$ $$\mathbb{Q}_{1,1}^{(c)}(E,c) = \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z478}}$ $$\mathbb{Q}_{1,2}^{(c)}(E,c) = -\frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z479}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$$

$\boxed{\text{z480}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$$

$\boxed{\text{z481}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,1b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z482}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,1b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z483}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,1c) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{14}$$

$\boxed{\text{z484}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,1c) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{14}$$

$\boxed{\text{z485}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,1d) = \frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_2)\mathbb{M}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z486}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,1d) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_2)\mathbb{M}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z487}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z488}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z489}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,2b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$$

$\boxed{\text{z490}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,2b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$$

$\boxed{\text{z491}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,2c) = \frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z492}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,2c) = -\frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z493}}$ $$\mathbb{Q}_{3,1}^{(c)}(E,1) = -\frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{30} - \frac{2\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{5}$$

$\boxed{\text{z494}}$ $$\mathbb{Q}_{3,2}^{(c)}(E,1) = \frac{2\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{5}$$

$\boxed{\text{z495}}$ $$\mathbb{Q}_{3,1}^{(c)}(E,2) = \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{6}$$

$\boxed{\text{z496}}$ $$\mathbb{Q}_{3,2}^{(c)}(E,2) = -\frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z497}}$ $$\mathbb{Q}_{4,1}^{(c)}(E,1) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{7}$$

$\boxed{\text{z498}}$ $$\mathbb{Q}_{4,2}^{(c)}(E,1) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{7}$$

$\boxed{\text{z499}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(E,a) = \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{10}$$

$\boxed{\text{z500}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(E,a) = \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{10}$$

$\boxed{\text{z501}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(E,b) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z502}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(E,b) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z503}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z504}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z505}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{14}$$

$\boxed{\text{z506}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1b) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{14}$$

$\boxed{\text{z507}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1c) = \frac{\sqrt{130}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{26} - \frac{\sqrt{130}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{26} - \frac{\sqrt{78}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{M}_{1,2}^{(b)}(E)}{26}$$

$\boxed{\text{z508}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1c) = -\frac{\sqrt{130}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{26} - \frac{\sqrt{130}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{26} + \frac{\sqrt{78}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{M}_{1,1}^{(b)}(E)}{26}$$

$\boxed{\text{z509}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1d) = \frac{\sqrt{2}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{M}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z510}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1d) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{M}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z511}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z512}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z513}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2b) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$$

$\boxed{\text{z514}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2b) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$$

$\boxed{\text{z515}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2c) = -\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{8} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{8} + \frac{\sqrt{15}\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{M}_{1,1}^{(b)}(E)}{8} + \frac{\sqrt{15}\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{M}_{1,2}^{(b)}(E)}{8}$$

$\boxed{\text{z516}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2c) = \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{8} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{8} + \frac{\sqrt{15}\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{M}_{1,2}^{(b)}(E)}{8} - \frac{\sqrt{15}\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{M}_{1,1}^{(b)}(E)}{8}$$

$\boxed{\text{z517}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2d) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z518}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2d) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z519}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,1a) = -\frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{30} - \frac{2\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{5}$$

$\boxed{\text{z520}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,1a) = \frac{2\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{5}$$

$\boxed{\text{z521}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{12} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{12} - \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z522}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z523}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,2a) = \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{6}$$

$\boxed{\text{z524}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,2a) = -\frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z525}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,2b) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{12} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{12} - \frac{\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{4} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,2}^{(b)}(E)}{4}$$

$\boxed{\text{z526}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,2b) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{12} + \frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{4} - \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,1}^{(b)}(E)}{4}$$

$\boxed{\text{z527}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E,1a) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{7}$$

$\boxed{\text{z528}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E,1a) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{7}$$

$\boxed{\text{z529}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E,1b) = \frac{\sqrt{39}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{26} - \frac{\sqrt{39}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{26} + \frac{\sqrt{65}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{M}_{1,2}^{(b)}(E)}{13}$$

$\boxed{\text{z530}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{39}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{26} - \frac{\sqrt{39}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{26} - \frac{\sqrt{65}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{M}_{1,1}^{(b)}(E)}{13}$$

$\boxed{\text{z531}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E,2) = \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{M}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{M}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z532}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E,2) = \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{M}_{1,2}^{(b)}(E)}{2} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{M}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z533}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E,3) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{8} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{8} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{M}_{1,1}^{(b)}(E)}{8} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{M}_{1,2}^{(b)}(E)}{8}$$

$\boxed{\text{z534}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E,3) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{8} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{8} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{M}_{1,2}^{(b)}(E)}{8} - \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{M}_{1,1}^{(b)}(E)}{8}$$

$\boxed{\text{z535}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(E,a) = \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z536}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(E,a) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z537}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(E,b) = \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{10}$$

$\boxed{\text{z538}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(E,b) = \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{10}$$

$\boxed{\text{z539}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,1a) = -\frac{\sqrt{2}\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$$

$\boxed{\text{z540}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,1a) = \frac{\sqrt{2}\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$$

$\boxed{\text{z541}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,1b) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z542}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,1b) = \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z543}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,1c) = \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{10} + \frac{\sqrt{30}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{M}_{1,1}^{(b)}(E)}{10}$$

$\boxed{\text{z544}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,1c) = -\frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{10} + \frac{\sqrt{30}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{M}_{1,2}^{(b)}(E)}{10}$$

$\boxed{\text{z545}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,2a) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$$

$\boxed{\text{z546}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,2a) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$$

$\boxed{\text{z547}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E,2b) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z548}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E,2b) = \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z549}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E,2c) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{2} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z550}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E,2c) = \frac{\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{M}_{1,2}^{(b)}(E)}{2} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z551}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E,1) = -\frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{30} - \frac{2\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{5}$

$\boxed{\text{z552}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E,1) = \frac{2\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{5}$

$\boxed{\text{z553}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E,2) = \frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6}$

$\boxed{\text{z554}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E,2) = -\frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{6}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6}$

$\boxed{\text{z555}}$ $\mathbb{Q}_{1,1}^{(1,1;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z556}}$ $\mathbb{Q}_{1,2}^{(1,1;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z557}}$ $\mathbb{Q}_{1,1}^{(1,1;c)}(E,b) = \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z558}}$ $\mathbb{Q}_{1,2}^{(1,1;c)}(E,b) = -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z559}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,1}^{(b)}(E,1)}{2}$

$\boxed{\text{z560}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{2,2}^{(b)}(E,1)}{2}$

$\boxed{\text{z561}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E,1b) = \frac{\sqrt{2}\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z562}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E,1b) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z563}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E,2) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z564}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E,2) = -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{M}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{M}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z565}}$ $\mathbb{G}_{1,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z566}}$ $\mathbb{G}_{1,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z567}}$ $\mathbb{G}_{2,1}^{(c)}(E,1a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z568}}$ $\mathbb{G}_{2,2}^{(c)}(E,1a) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{6} + \frac{\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z569}}$ $\mathbb{G}_{2,1}^{(c)}(E,1b) = \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z570}}$ $\mathbb{G}_{2,2}^{(c)}(E,1b) = -\frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z571}}$ $\mathbb{G}_{2,1}^{(c)}(E,2a) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z572}}$ $\mathbb{G}_{2,2}^{(c)}(E,2a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6}$

$\boxed{\text{z573}}$ $\mathbb{G}_{2,1}^{(c)}(E,2b) = \frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z574}}$ $\mathbb{G}_{2,2}^{(c)}(E,2b) = -\frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z575}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(E) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z576}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(E) = \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z577}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,1a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z578}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,1a) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{6} + \frac{\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z579}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,1b) = \frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{42} + \frac{2\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{21} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{42} - \frac{\sqrt{14}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1,2}^{(b)}(E)}{14}$$

$\boxed{\text{z580}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,1b) = -\frac{2\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{21} - \frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{42} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{42} + \frac{\sqrt{14}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1,1}^{(b)}(E)}{14}$$

$\boxed{\text{z581}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,1c) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z582}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,1c) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z583}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,2a) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z584}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,2a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1}^{(b)}(A_2)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z585}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,2b) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{42} + \frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{42} + \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{42} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{14} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,2}^{(b)}(E)}{14}$$

$\boxed{\text{z586}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,2b) = \frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{42} - \frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{42} + \frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{42} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,1}^{(b)}(E)}{14}$$

$\boxed{\text{z587}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,2c) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z588}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,2c) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z589}}$ $\mathbb{G}_{3,1}^{(1,-1;c)}(E,1) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z590}}$ $\mathbb{G}_{3,2}^{(1,-1;c)}(E,1) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z591}}$ $\mathbb{G}_{3,1}^{(1,-1;c)}(E,2) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z592}}$ $\mathbb{G}_{3,2}^{(1,-1;c)}(E,2) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z593}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(E,1) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{28} + \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{28} + \frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{28} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1,2}^{(b)}(E)}{14}$

$\boxed{\text{z594}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(E,1) = -\frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{28} + \frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{28} + \frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{28} - \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1,1}^{(b)}(E)}{14}$

$\boxed{\text{z595}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(E,2) = \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z596}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(E,2) = \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{2} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z597}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(E,3) = \frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{14} - \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{28} - \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,2}^{(b)}(E)}{28}$

$\boxed{\text{z598}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(E,3) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{14} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{28} - \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,1}^{(b)}(E)}{28}$

$\boxed{\text{z599}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(E,a) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z600}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(E,a) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z601}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(E,b) = \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{M}_{1,1}^{(b)}(E)}{5}$

$\boxed{\text{z602}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(E,b) = -\frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{M}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{M}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{M}_{1,2}^{(b)}(E)}{5}$

$\boxed{\text{z603}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E,1a) = \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z604}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E,1a) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z605}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E,1b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z606}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E,1b) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z607}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E,2a) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z608}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E,2a) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z609}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E,2b) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z610}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E,2b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z611}}$ $$\mathbb{G}_{1,1}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z612}}$ $$\mathbb{G}_{1,2}^{(1,1;c)}(E) = \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z613}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E,1) = \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z614}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E,1) = -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z615}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E,2) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z616}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E,2) = -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

- `A;A_002_1` : `'A'-'A'` bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `3a@3b`

$\boxed{\text{z93}}$ $\mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z94}}$ $\mathbb{G}_0^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z95}}$ $\mathbb{G}_2^{(1,-1;c)}(A_1) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z221}}$ $\mathbb{Q}_1^{(1,-1;c)}(A_2) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z617}}$ $\mathbb{Q}_{1,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z618}}$ $\mathbb{Q}_{1,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z619}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(E) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z620}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(E) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z621}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E,1) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z622}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E,1) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z623}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E,2) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z624}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E,2) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

- `A;A_002_1` : `'A'-'A'` bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3a@3b`

$\boxed{\text{z96}}$ $\mathbb{Q}_0^{(c)}(A_1,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z97}}$ $\mathbb{Q}_0^{(c)}(A_1,b) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z98}}$ $\mathbb{Q}_2^{(c)}(A_1) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z99}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_1,a) = -\frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z100}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_1,b) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z101}}$ $\mathbb{Q}_0^{(1,0;c)}(A_1,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z102}}$ $\mathbb{Q}_0^{(1,0;c)}(A_1,b) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z103}}$ $\mathbb{Q}_2^{(1,0;c)}(A_1) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z104}}$ $\mathbb{G}_2^{(1,-1;c)}(A_1) = \mathbb{G}_2^{(1,-1;a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z105}}$ $\mathbb{G}_3^{(1,-1;c)}(A_1,a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z106}}$ $\mathbb{G}_3^{(1,-1;c)}(A_1,b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z107}}$ $\mathbb{G}_0^{(1,1;c)}(A_1) = \mathbb{G}_0^{(1,1;a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z222}}$ $\mathbb{Q}_1^{(c)}(A_2) = \mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z223}}$ $\mathbb{Q}_1^{(1,0;c)}(A_2) = \mathbb{Q}_1^{(1,0;a)}(A_2)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z224}}$ $\mathbb{G}_1^{(c)}(A_2, a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z225}}$ $\mathbb{G}_1^{(c)}(A_2, b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z226}}$ $\mathbb{G}_1^{(1,-1;c)}(A_2, a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z227}}$ $\mathbb{G}_1^{(1,-1;c)}(A_2, b) = \frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(1,-1;a)}(A_1)\mathbb{T}_1^{(b)}(A_2)}{5}$

$\boxed{\text{z228}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2, 1) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{5} + \frac{\sqrt{15}\mathbb{M}_2^{(1,-1;a)}(A_1)\mathbb{T}_1^{(b)}(A_2)}{5}$

$\boxed{\text{z229}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2, 2a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z230}}$ $\mathbb{G}_3^{(1,-1;c)}(A_2, 2b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z231}}$ $\mathbb{G}_1^{(1,0;c)}(A_2, a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z232}}$ $\mathbb{G}_1^{(1,0;c)}(A_2, b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z233}}$ $\mathbb{G}_1^{(1,1;c)}(A_2) = \mathbb{M}_0^{(1,1;a)}(A_1)\mathbb{T}_1^{(b)}(A_2)$

$\boxed{\text{z625}}$ $\mathbb{Q}_{1,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_0^{(b)}(A_1)}{2}$

$\boxed{\text{z626}}$ $\mathbb{Q}_{1,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_0^{(b)}(A_1)}{2}$

$\boxed{\text{z627}}$ $\mathbb{Q}_{2,1}^{(c)}(E, 1a) = \frac{\sqrt{2}\mathbb{Q}_1^{(a)}(A_2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z628}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,1a) = -\frac{\sqrt{2}\mathbb{Q}_{1}^{(a)}(A_2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z629}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,1b) = \frac{\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z630}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,1b) = -\frac{\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z631}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,2a) = \frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z632}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,2a) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z633}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,2b) = \frac{\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z634}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,2b) = -\frac{\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z635}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1a) = \frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{10} + \frac{\sqrt{30}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{10}$$

$\boxed{\text{z636}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1a) = -\frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{10} + \frac{\sqrt{30}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{10}$$

$\boxed{\text{z637}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1b) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z638}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z639}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2a) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z640}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2a) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2} + \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z641}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2b) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z642}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2b) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z643}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z644}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z645}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,1a) = \frac{\sqrt{2}\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z646}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,1a) = -\frac{\sqrt{2}\mathbb{Q}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z647}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,1b) = \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z648}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,1b) = -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z649}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,2a) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z650}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,2a) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z651}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E,2b) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z652}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E,2b) = -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z653}}$ $$\mathbb{G}_{1,1}^{(c)}(E) = \frac{\mathbb{T}_{1,2}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z654}}$ $$\mathbb{G}_{1,2}^{(c)}(E) = -\frac{\mathbb{T}_{1,1}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{T}_{1}^{(a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z655}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(E,a) = \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{5}$$

$\boxed{\text{z656}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(E,a) = -\frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{G}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{5}$$

$\boxed{\text{z657}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(E,b) = \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{10}$$

$\boxed{\text{z658}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(E,b) = \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{10}$$

$\boxed{\text{z659}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,1) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z660}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,1) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z661}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E,2) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z662}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E,2) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z663}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(E,1) = -\frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{30} - \frac{2\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{5}$$

$\boxed{\text{z664}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(E,1) = \frac{2\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{M}_{2}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{5}$$

$\boxed{\text{z665}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(E,2) = \frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{2,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6}$$

$\boxed{\text{z666}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(E,2) = -\frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{6}\mathbb{M}_{2,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z667}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(E) = \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z668}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(E) = -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{T}_{1}^{(1,0;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z669}}$ $$\mathbb{G}_{1,1}^{(1,1;c)}(E,a) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z670}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(E,a) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z671}}$ $\mathbb{G}_{1,1}^{(1,1;c)}(E,b) = \frac{\sqrt{2}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z672}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(E,b) = \frac{\sqrt{2}\mathbb{M}_{0}^{(1,1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

- `A;A_002_1` : `'A'-'A'` bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3a@3b`

$\boxed{\text{z108}}$ $\mathbb{Q}_{0}^{(c)}(A_1) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z109}}$ $\mathbb{Q}_{2}^{(c)}(A_1) = \mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z110}}$ $\mathbb{Q}_{3}^{(c)}(A_1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z111}}$ $\mathbb{Q}_{2}^{(1,-1;c)}(A_1) = \mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z112}}$ $\mathbb{Q}_{3}^{(1,-1;c)}(A_1,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z113}}$ $\mathbb{Q}_{3}^{(1,-1;c)}(A_1,b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{4} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{4} + \frac{\sqrt{3}\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1}^{(b)}(A_2)}{2}$

$\boxed{\text{z114}}$ $\mathbb{Q}_{3}^{(1,0;c)}(A_1) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z115}}$ $\mathbb{Q}_{0}^{(1,1;c)}(A_1) = \mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z116}}$ $\mathbb{G}_{0}^{(c)}(A_1) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z117}}$ $\mathbb{G}_{2}^{(c)}(A_1,a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z118}}$ $$\mathbb{G}_2^{(c)}(A_1,b) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z119}}$ $$\mathbb{G}_0^{(1,-1;c)}(A_1) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z120}}$ $$\mathbb{G}_2^{(1,-1;c)}(A_1,a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z121}}$ $$\mathbb{G}_2^{(1,-1;c)}(A_1,b) = \frac{\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{7} + \frac{\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{7} + \frac{\sqrt{21}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1}^{(b)}(A_2)}{7}$$

$\boxed{\text{z122}}$ $$\mathbb{G}_2^{(1,-1;c)}(A_1,c) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z123}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_1,1) = -\frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{14} + \frac{2\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1}^{(b)}(A_2)}{7}$$

$\boxed{\text{z124}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_1,2) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{4} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{4} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1}^{(b)}(A_2)}{2}$$

$\boxed{\text{z125}}$ $$\mathbb{G}_0^{(1,0;c)}(A_1) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z126}}$ $$\mathbb{G}_2^{(1,0;c)}(A_1) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z127}}$ $$\mathbb{G}_0^{(1,1;c)}(A_1) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z128}}$ $$\mathbb{G}_2^{(1,1;c)}(A_1) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$$

$\boxed{\text{z234}}$ $$\mathbb{Q}_1^{(c)}(A_2,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z235}}$ $$\mathbb{Q}_1^{(c)}(A_2,b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z236}}$ $$\mathbb{Q}_3^{(c)}(A_2,2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z237}}$ $\mathbb{Q}_{1}^{(1,-1;c)}(A_2,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z238}}$ $\mathbb{Q}_{1}^{(1,-1;c)}(A_2,b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z239}}$ $\mathbb{Q}_{3}^{(1,-1;c)}(A_2,1) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z240}}$ $\mathbb{Q}_{3}^{(1,-1;c)}(A_2,2a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z241}}$ $\mathbb{Q}_{3}^{(1,-1;c)}(A_2,2b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{4} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{4} - \frac{\sqrt{3}\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1}^{(b)}(A_2)}{2}$

$\boxed{\text{z242}}$ $\mathbb{Q}_{1}^{(1,0;c)}(A_2,a) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z243}}$ $\mathbb{Q}_{1}^{(1,0;c)}(A_2,b) = \frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{10} + \frac{\sqrt{10}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1}^{(b)}(A_2)}{5}$

$\boxed{\text{z244}}$ $\mathbb{Q}_{3}^{(1,0;c)}(A_2,1) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{5} + \frac{\sqrt{15}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1}^{(b)}(A_2)}{5}$

$\boxed{\text{z245}}$ $\mathbb{Q}_{3}^{(1,0;c)}(A_2,2) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z246}}$ $\mathbb{Q}_{1}^{(1,1;c)}(A_2) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z247}}$ $\mathbb{G}_{4}^{(1,-1;c)}(A_2) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{4} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{4} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1}^{(b)}(A_2)}{2}$

$\boxed{\text{z248}}$ $\mathbb{G}_{1}^{(1,0;c)}(A_2) = \mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{0}^{(b)}(A_1)$

$\boxed{\text{z673}}$ $\mathbb{Q}_{1,1}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z674}}$ $\mathbb{Q}_{1,2}^{(c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z675}}$ $$\mathbb{Q}_{1,1}^{(c)}(E,b) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{14}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{14}$$

$\boxed{\text{z676}}$ $$\mathbb{Q}_{1,2}^{(c)}(E,b) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{14}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{14}$$

$\boxed{\text{z677}}$ $$\mathbb{Q}_{1,1}^{(c)}(E,c) = \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z678}}$ $$\mathbb{Q}_{1,2}^{(c)}(E,c) = -\frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z679}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z680}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,1) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z681}}$ $$\mathbb{Q}_{2,1}^{(c)}(E,2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z682}}$ $$\mathbb{Q}_{2,2}^{(c)}(E,2) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z683}}$ $$\mathbb{Q}_{3,1}^{(c)}(E,1) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{7}$$

$\boxed{\text{z684}}$ $$\mathbb{Q}_{3,2}^{(c)}(E,1) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{7}$$

$\boxed{\text{z685}}$ $$\mathbb{Q}_{3,1}^{(c)}(E,2) = \frac{\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z686}}$ $$\mathbb{Q}_{3,2}^{(c)}(E,2) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z687}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(E,a) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{14}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{14}$$

$\boxed{\text{z688}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(E,a) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{14}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{14}$$

$\boxed{\text{z689}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(E,b) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z690}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(E,b) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,-1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z691}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z692}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,1) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z693}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E,2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z694}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E,2) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$$

$\boxed{\text{z695}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,1a) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{7}$$

$\boxed{\text{z696}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,1a) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} + \frac{\sqrt{21}\mathbb{Q}_{2}^{(1,-1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{7}$$

$\boxed{\text{z697}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{12} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{12} - \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z698}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,1b) = -\frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,1)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z699}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,2a) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z700}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,2a) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z701}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E,2b) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{12} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{12} - \frac{\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{4} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,2}^{(b)}(E)}{4}$$

$\boxed{\text{z702}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E,2b) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{12} + \frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{4} - \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,1}^{(b)}(E)}{4}$$

$\boxed{\text{z703}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(E,a) = -\frac{\sqrt{2}\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z704}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(E,a) = \frac{\sqrt{2}\mathbb{G}_{1}^{(1,0;a)}(A_2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z705}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(E,b) = \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{10}$$

$\boxed{\text{z706}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(E,b) = \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{10} - \frac{\sqrt{5}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{10}$$

$\boxed{\text{z707}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E,1) = -\frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{30} - \frac{2\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{5}$$

$\boxed{\text{z708}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E,1) = \frac{2\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{15} + \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{30} + \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{30} + \frac{\sqrt{5}\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{5}$$

$\boxed{\text{z709}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E,2) = \frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6}$$

$\boxed{\text{z710}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E,2) = -\frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{6}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6}$$

$\boxed{\text{z711}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z712}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(E,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z713}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(E,b) = \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z714}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(E,b) = -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z715}}$ $$\mathbb{G}_{2,1}^{(c)}(E,1) = \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$\boxed{\text{z716}}$ $$\mathbb{G}_{2,2}^{(c)}(E,1) = -\frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$\boxed{\text{z717}}$ $$\mathbb{G}^{(c)}_{2,1}(E,2) = \frac{\mathbb{M}^{(a)}_{1,1}(E)\mathbb{T}^{(b)}_{1,1}(E)}{2} - \frac{\mathbb{M}^{(a)}_{1,2}(E)\mathbb{T}^{(b)}_{1,2}(E)}{2}$$

$\boxed{\text{z718}}$ $$\mathbb{G}^{(c)}_{2,2}(E,2) = -\frac{\mathbb{M}^{(a)}_{1,1}(E)\mathbb{T}^{(b)}_{1,2}(E)}{2} - \frac{\mathbb{M}^{(a)}_{1,2}(E)\mathbb{T}^{(b)}_{1,1}(E)}{2}$$

$\boxed{\text{z719}}$ $$\mathbb{G}^{(1,-1;c)}_{2,1}(E,1a) = \frac{\sqrt{210}\mathbb{M}^{(1,-1;a)}_{3,1}(E,2)\mathbb{T}^{(b)}_{1,1}(E)}{42} + \frac{2\sqrt{21}\mathbb{M}^{(1,-1;a)}_{3,2}(E,1)\mathbb{T}^{(b)}_{1}(A_2)}{21} - \frac{\sqrt{210}\mathbb{M}^{(1,-1;a)}_{3,2}(E,2)\mathbb{T}^{(b)}_{1,2}(E)}{42} - \frac{\sqrt{14}\mathbb{M}^{(1,-1;a)}_{3}(A_2,1)\mathbb{T}^{(b)}_{1,2}(E)}{14}$$

$\boxed{\text{z720}}$ $$\mathbb{G}^{(1,-1;c)}_{2,2}(E,1a) = -\frac{2\sqrt{21}\mathbb{M}^{(1,-1;a)}_{3,1}(E,1)\mathbb{T}^{(b)}_{1}(A_2)}{21} - \frac{\sqrt{210}\mathbb{M}^{(1,-1;a)}_{3,1}(E,2)\mathbb{T}^{(b)}_{1,2}(E)}{42} - \frac{\sqrt{210}\mathbb{M}^{(1,-1;a)}_{3,2}(E,2)\mathbb{T}^{(b)}_{1,1}(E)}{42} + \frac{\sqrt{14}\mathbb{M}^{(1,-1;a)}_{3}(A_2,1)\mathbb{T}^{(b)}_{1,1}(E)}{14}$$

$\boxed{\text{z721}}$ $$\mathbb{G}^{(1,-1;c)}_{2,1}(E,1b) = \frac{\mathbb{M}^{(1,-1;a)}_{1,2}(E)\mathbb{T}^{(b)}_{1}(A_2)}{2} + \frac{\mathbb{M}^{(1,-1;a)}_{1}(A_2)\mathbb{T}^{(b)}_{1,2}(E)}{2}$$

$\boxed{\text{z722}}$ $$\mathbb{G}^{(1,-1;c)}_{2,2}(E,1b) = -\frac{\mathbb{M}^{(1,-1;a)}_{1,1}(E)\mathbb{T}^{(b)}_{1}(A_2)}{2} - \frac{\mathbb{M}^{(1,-1;a)}_{1}(A_2)\mathbb{T}^{(b)}_{1,1}(E)}{2}$$

$\boxed{\text{z723}}$ $$\mathbb{G}^{(1,-1;c)}_{2,1}(E,2a) = -\frac{\sqrt{21}\mathbb{M}^{(1,-1;a)}_{3,1}(E,1)\mathbb{T}^{(b)}_{1,1}(E)}{42} + \frac{\sqrt{21}\mathbb{M}^{(1,-1;a)}_{3,2}(E,1)\mathbb{T}^{(b)}_{1,2}(E)}{42} + \frac{\sqrt{210}\mathbb{M}^{(1,-1;a)}_{3,2}(E,2)\mathbb{T}^{(b)}_{1}(A_2)}{42} + \frac{\sqrt{35}\mathbb{M}^{(1,-1;a)}_{3}(A_1)\mathbb{T}^{(b)}_{1,1}(E)}{14} + \frac{\sqrt{35}\mathbb{M}^{(1,-1;a)}_{3}(A_2,2)\mathbb{T}^{(b)}_{1,2}(E)}{14}$$

$\boxed{\text{z724}}$ $$\mathbb{G}^{(1,-1;c)}_{2,2}(E,2a) = \frac{\sqrt{21}\mathbb{M}^{(1,-1;a)}_{3,1}(E,1)\mathbb{T}^{(b)}_{1,2}(E)}{42} - \frac{\sqrt{210}\mathbb{M}^{(1,-1;a)}_{3,1}(E,2)\mathbb{T}^{(b)}_{1}(A_2)}{42} + \frac{\sqrt{21}\mathbb{M}^{(1,-1;a)}_{3,2}(E,1)\mathbb{T}^{(b)}_{1,1}(E)}{42} + \frac{\sqrt{35}\mathbb{M}^{(1,-1;a)}_{3}(A_1)\mathbb{T}^{(b)}_{1,2}(E)}{14} - \frac{\sqrt{35}\mathbb{M}^{(1,-1;a)}_{3}(A_2,2)\mathbb{T}^{(b)}_{1,1}(E)}{14}$$

$\boxed{\text{z725}}$ $$\mathbb{G}^{(1,-1;c)}_{2,1}(E,2b) = \frac{\mathbb{M}^{(1,-1;a)}_{1,1}(E)\mathbb{T}^{(b)}_{1,1}(E)}{2} - \frac{\mathbb{M}^{(1,-1;a)}_{1,2}(E)\mathbb{T}^{(b)}_{1,2}(E)}{2}$$

$\boxed{\text{z726}}$ $$\mathbb{G}^{(1,-1;c)}_{2,2}(E,2b) = -\frac{\mathbb{M}^{(1,-1;a)}_{1,1}(E)\mathbb{T}^{(b)}_{1,2}(E)}{2} - \frac{\mathbb{M}^{(1,-1;a)}_{1,2}(E)\mathbb{T}^{(b)}_{1,1}(E)}{2}$$

$\boxed{\text{z727}}$ $$\mathbb{G}^{(1,-1;c)}_{4,1}(E,1) = -\frac{\sqrt{21}\mathbb{M}^{(1,-1;a)}_{3,1}(E,2)\mathbb{T}^{(b)}_{1,1}(E)}{28} + \frac{\sqrt{210}\mathbb{M}^{(1,-1;a)}_{3,2}(E,1)\mathbb{T}^{(b)}_{1}(A_2)}{28} + \frac{\sqrt{21}\mathbb{M}^{(1,-1;a)}_{3,2}(E,2)\mathbb{T}^{(b)}_{1,2}(E)}{28} + \frac{\sqrt{35}\mathbb{M}^{(1,-1;a)}_{3}(A_2,1)\mathbb{T}^{(b)}_{1,2}(E)}{14}$$

$\boxed{\text{z728}}$ $$\mathbb{G}^{(1,-1;c)}_{4,2}(E,1) = -\frac{\sqrt{210}\mathbb{M}^{(1,-1;a)}_{3,1}(E,1)\mathbb{T}^{(b)}_{1}(A_2)}{28} + \frac{\sqrt{21}\mathbb{M}^{(1,-1;a)}_{3,1}(E,2)\mathbb{T}^{(b)}_{1,2}(E)}{28} + \frac{\sqrt{21}\mathbb{M}^{(1,-1;a)}_{3,2}(E,2)\mathbb{T}^{(b)}_{1,1}(E)}{28} - \frac{\sqrt{35}\mathbb{M}^{(1,-1;a)}_{3}(A_2,1)\mathbb{T}^{(b)}_{1,1}(E)}{14}$$

$\boxed{\text{z729}}$ $$\mathbb{G}^{(1,-1;c)}_{4,1}(E,2) = \frac{\mathbb{M}^{(1,-1;a)}_{3}(A_1)\mathbb{T}^{(b)}_{1,1}(E)}{2} - \frac{\mathbb{M}^{(1,-1;a)}_{3}(A_2,2)\mathbb{T}^{(b)}_{1,2}(E)}{2}$$

$\boxed{\text{z730}}$ $$\mathbb{G}^{(1,-1;c)}_{4,2}(E,2) = \frac{\mathbb{M}^{(1,-1;a)}_{3}(A_1)\mathbb{T}^{(b)}_{1,2}(E)}{2} + \frac{\mathbb{M}^{(1,-1;a)}_{3}(A_2,2)\mathbb{T}^{(b)}_{1,1}(E)}{2}$$

$\boxed{\text{z731}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(E,3) = \frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{14} - \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{28} - \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,2}^{(b)}(E)}{28}$

$\boxed{\text{z732}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(E,3) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{14} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{28} - \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_2,2)\mathbb{T}_{1,1}^{(b)}(E)}{28}$

$\boxed{\text{z733}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z734}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(E) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{0}^{(b)}(A_1)}{2}$

$\boxed{\text{z735}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(E,1) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z736}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(E,1) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1}^{(b)}(A_2)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\mathbb{T}_{2}^{(1,0;a)}(A_1)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z737}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(E,2a) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z738}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(E,2a) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z739}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(E,2b) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{3}$

$\boxed{\text{z740}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(E,2b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,1)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E,2)\mathbb{T}_{1}^{(b)}(A_2)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E,1)\mathbb{T}_{1,1}^{(b)}(E)}{6}$

$\boxed{\text{z741}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(E,1) = \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} + \frac{\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z742}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E,1) = -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1}^{(b)}(A_2)}{2} - \frac{\mathbb{M}_{1}^{(1,1;a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

$\boxed{\text{z743}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(E,2) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$

$\boxed{\text{z744}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E,2) = -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$

Atomic SAMB

- bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
- ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$

$\boxed{\text{x1}}$ $\mathbb{Q}_0^{(a)}(A_1) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$

$\boxed{\text{x2}}$ $\mathbb{M}_1^{(1,-1;a)}(A_2) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$

$\boxed{\text{x3}}$ $\mathbb{M}_{1,1}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

$\boxed{\text{x4}}$ $\mathbb{M}_{1,2}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

- bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
- ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$

$\boxed{\text{x5}}$ $\mathbb{Q}_1^{(a)}(A_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$

$\boxed{\text{x6}}$ $\mathbb{Q}_{1,1}^{(a)}(E) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x7}}$ $\mathbb{Q}_{1,2}^{(a)}(E) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x8}}$ $\mathbb{Q}_1^{(1,0;a)}(A_2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x9}}$ $\mathbb{Q}_{1,1}^{(1,0;a)}(E) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x10}}$ $\mathbb{Q}_{1,2}^{(1,0;a)}(E) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x11}}$ $\mathbb{G}_{2}^{(1,-1;a)}(A_1) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \end{bmatrix}$

$\boxed{\text{x12}}$ $\mathbb{G}_{2,1}^{(1,-1;a)}(E,1) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x13}}$ $\mathbb{G}_{2,2}^{(1,-1;a)}(E,1) = \begin{bmatrix} -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x14}}$ $\mathbb{G}_{2,1}^{(1,-1;a)}(E,2) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x15}}$ $\mathbb{G}_{2,2}^{(1,-1;a)}(E,2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x16}}$ $\mathbb{G}_{0}^{(1,1;a)}(A_1) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$

$\boxed{\text{x17}}$ $\mathbb{M}_{2}^{(1,-1;a)}(A_1) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{6} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$

$\boxed{\text{x18}}$ $\mathbb{M}_{2,1}^{(1,-1;a)}(E,1) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x19}}$ $\mathbb{M}_{2,2}^{(1,-1;a)}(E,1) = \begin{bmatrix} -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x20}}$ $\mathbb{M}_{2,1}^{(1,-1;a)}(E,2) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x21}}$ $\mathbb{M}_{2,2}^{(1,-1;a)}(E,2) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x22}}$ $\mathbb{M}_{0}^{(1,1;a)}(A_1) = \begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$

$\boxed{\text{x23}}$ $\mathbb{T}_{1}^{(a)}(A_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$

$\boxed{\text{x24}}$ $\mathbb{T}_{1,1}^{(a)}(E) = \begin{bmatrix} \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x25}}$ $\mathbb{T}_{1,2}^{(a)}(E) = \begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x26}}$ $\mathbb{T}_{1}^{(1,0;a)}(A_2) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x27}}$ $\mathbb{T}_{1,1}^{(1,0;a)}(E) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x28}}$ $\mathbb{T}_{1,2}^{(1,0;a)}(E) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

- bra: $\langle p_x, \uparrow|$, $\langle p_x, \downarrow|$, $\langle p_y, \uparrow|$, $\langle p_y, \downarrow|$, $\langle p_z, \uparrow|$, $\langle p_z, \downarrow|$
- ket: $|p_x, \uparrow\rangle$, $|p_x, \downarrow\rangle$, $|p_y, \uparrow\rangle$, $|p_y, \downarrow\rangle$, $|p_z, \uparrow\rangle$, $|p_z, \downarrow\rangle$

$\boxed{\text{x29}}$ $\mathbb{Q}_0^{(a)}(A_1) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$

$\boxed{\text{x30}}$ $\mathbb{Q}_2^{(a)}(A_1) = \begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$

$\boxed{\text{x31}}$ $\mathbb{Q}_{2,1}^{(a)}(E,1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x32}}$ $\mathbb{Q}_{2,2}^{(a)}(E,1) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x33}}$ $\mathbb{Q}_{2,1}^{(a)}(E,2) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x34}}$ $\mathbb{Q}_{2,2}^{(a)}(E,2) = \begin{bmatrix} 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x35}}$ $\mathbb{Q}_{2}^{(1,-1;a)}(A_1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x36}}$ $\mathbb{Q}_{2,1}^{(1,-1;a)}(E,1) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x37}}$ $\mathbb{Q}_{2,2}^{(1,-1;a)}(E,1) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x38}}$ $\mathbb{Q}_{2,1}^{(1,-1;a)}(E,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x39}}$ $\mathbb{Q}_{2,2}^{(1,-1;a)}(E,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x40}}$ $\mathbb{Q}_{0}^{(1,1;a)}(A_1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x41}}$ $\mathbb{G}_1^{(1,0;a)}(A_2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x42}}$ $\mathbb{G}_{1,1}^{(1,0;a)}(E) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x43}}$ $\mathbb{G}_{1,2}^{(1,0;a)}(E) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x44}}$ $\mathbb{M}_1^{(a)}(A_2) = \begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x45}}$ $$\mathbb{M}_{1,1}^{(a)}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x46}}$ $$\mathbb{M}_{1,2}^{(a)}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x47}}$ $$\mathbb{M}_{3}^{(1,-1;a)}(A_1) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x48}}$ $$\mathbb{M}_{1}^{(1,-1;a)}(A_2) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$$

$\boxed{\text{x49}}$ $\mathbb{M}_3^{(1,-1;a)}(A_2,1) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$

$\boxed{\text{x50}}$ $\mathbb{M}_3^{(1,-1;a)}(A_2,2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x51}}$ $\mathbb{M}_{1,1}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$

$\boxed{\text{x52}}$ $\mathbb{M}_{1,2}^{(1,-1;a)}(E) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$

$\boxed{\text{x53}}$ $\mathbb{M}_{3,1}^{(1,-1;a)}(E,1) = \begin{bmatrix} 0 & -\frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{60} & \frac{\sqrt{30}}{15} & 0 \\ -\frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{30}}{15} \\ 0 & \frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} \\ 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & \frac{\sqrt{30}}{15} & 0 \end{bmatrix}$

$\boxed{\text{x54}}$ $\mathbb{M}_{3,2}^{(1,-1;a)}(E,1) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{60} & 0 & \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ -\frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \\ 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & -\frac{\sqrt{30}i}{15} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{15} & \frac{\sqrt{30}i}{15} & 0 \end{bmatrix}$

$\boxed{\text{x55}}$ $\mathbb{M}_{3,1}^{(1,-1;a)}(E,2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x56}}$ $\mathbb{M}_{3,2}^{(1,-1;a)}(E,2) = \begin{bmatrix} \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x57}}$ $\mathbb{M}_1^{(1,1;a)}(A_2) = \begin{bmatrix} -\frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & \frac{\sqrt{30}}{30} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{30} & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \end{bmatrix}$

$\boxed{\text{x58}}$ $\mathbb{M}_{1,1}^{(1,1;a)}(E) = \begin{bmatrix} 0 & \frac{\sqrt{30}}{15} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{15} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 \end{bmatrix}$

$\boxed{\text{x59}}$ $\mathbb{M}_{1,2}^{(1,1;a)}(E) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{30} & 0 \end{bmatrix}$

$\boxed{\text{x60}}$ $\mathbb{T}_2^{(1,0;a)}(A_1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x61}}$ $\mathbb{T}_{2,1}^{(1,0;a)}(E,1) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$

$\boxed{\text{x62}}$ $\mathbb{T}_{2,2}^{(1,0;a)}(E,1) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$

$\boxed{\text{x63}}$ $\mathbb{T}_{2,1}^{(1,0;a)}(E,2) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x64}}$ $\mathbb{T}_{2,2}^{(1,0;a)}(E,2) = \begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$

Cluster SAMB

- Site cluster

** Wyckoff: `6c`

$\boxed{\text{y1}}$ $\mathbb{Q}_0^{(s)}(A_1) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y2}}$ $\mathbb{Q}_1^{(s)}(A_2) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y3}}$ $\mathbb{Q}_{1,1}^{(s)}(E) = \left[\frac{5\sqrt{21}}{42}, -\frac{2\sqrt{21}}{21}, -\frac{\sqrt{21}}{42}, -\frac{\sqrt{21}}{42}, \frac{5\sqrt{21}}{42}, -\frac{2\sqrt{21}}{21}\right]$

$\boxed{\text{y4}}$ $\mathbb{Q}_{1,2}^{(s)}(E) = \left[\frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7}, -\frac{3\sqrt{7}}{14}, \frac{3\sqrt{7}}{14}, -\frac{\sqrt{7}}{14}, -\frac{\sqrt{7}}{7}\right]$

$\boxed{\text{y5}}$ $\mathbb{Q}_{2,1}^{(s)}(E,1) = \left[\frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7}, -\frac{3\sqrt{7}}{14}, -\frac{3\sqrt{7}}{14}, \frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7}\right]$

$\boxed{\text{y6}}$ $\mathbb{Q}_{2,2}^{(s)}(E,1) = \left[-\frac{5\sqrt{21}}{42}, \frac{2\sqrt{21}}{21}, \frac{\sqrt{21}}{42}, -\frac{\sqrt{21}}{42}, \frac{5\sqrt{21}}{42}, -\frac{2\sqrt{21}}{21}\right]$

- Bond cluster

** Wyckoff: `3a@3b`

$\boxed{\text{y7}}$ $\mathbb{Q}_0^{(s)}(A_1) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

$\boxed{\text{y8}}$ $\mathbb{T}_1^{(s)}(A_2) = \left[\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}\right]$

$\boxed{\text{y9}}$ $\mathbb{Q}_{1,1}^{(s)}(E) = \left[\frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y10}}$ $\mathbb{Q}_{1,2}^{(s)}(E) = \left[0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right]$

$\boxed{\text{y11}}$ $\mathbb{T}_{1,1}^{(s)}(E) = \left[0,\ -\frac{\sqrt{2}i}{2},\ \frac{\sqrt{2}i}{2}\right]$

$\boxed{\text{y12}}$ $\mathbb{T}_{1,2}^{(s)}(E) = \left[\frac{\sqrt{6}i}{3},\ -\frac{\sqrt{6}i}{6},\ -\frac{\sqrt{6}i}{6}\right]$

** Wyckoff: `6a@6c`

$\boxed{\text{y13}}$ $\mathbb{Q}_{0}^{(s)}(A_1) = \left[\frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y14}}$ $\mathbb{T}_{0}^{(s)}(A_1) = \left[\frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6}\right]$

$\boxed{\text{y15}}$ $\mathbb{Q}_{1}^{(s)}(A_2) = \left[\frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ \frac{\sqrt{6}}{6},\ -\frac{\sqrt{6}}{6},\ -\frac{\sqrt{6}}{6},\ -\frac{\sqrt{6}}{6}\right]$

$\boxed{\text{y16}}$ $\mathbb{T}_{1}^{(s)}(A_2) = \left[\frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ \frac{\sqrt{6}i}{6},\ -\frac{\sqrt{6}i}{6},\ -\frac{\sqrt{6}i}{6},\ -\frac{\sqrt{6}i}{6}\right]$

$\boxed{\text{y17}}$ $\mathbb{Q}_{1,1}^{(s)}(E) = \left[\frac{5\sqrt{21}}{42},\ -\frac{2\sqrt{21}}{21},\ -\frac{\sqrt{21}}{42},\ -\frac{\sqrt{21}}{42},\ \frac{5\sqrt{21}}{42},\ -\frac{2\sqrt{21}}{21}\right]$

$\boxed{\text{y18}}$ $\mathbb{Q}_{1,2}^{(s)}(E) = \left[\frac{\sqrt{7}}{14},\ \frac{\sqrt{7}}{7},\ -\frac{3\sqrt{7}}{14},\ \frac{3\sqrt{7}}{14},\ -\frac{\sqrt{7}}{14},\ -\frac{\sqrt{7}}{7}\right]$

$\boxed{\text{y19}}$ $\mathbb{T}_{1,1}^{(s)}(E) = \left[\frac{5\sqrt{21}i}{42},\ -\frac{2\sqrt{21}i}{21},\ -\frac{\sqrt{21}i}{42},\ -\frac{\sqrt{21}i}{42},\ \frac{5\sqrt{21}i}{42},\ -\frac{2\sqrt{21}i}{21}\right]$

$\boxed{\text{y20}}$ $\mathbb{T}_{1,2}^{(s)}(E) = \left[\frac{\sqrt{7}i}{14},\ \frac{\sqrt{7}i}{7},\ -\frac{3\sqrt{7}i}{14},\ \frac{3\sqrt{7}i}{14},\ -\frac{\sqrt{7}i}{14},\ -\frac{\sqrt{7}i}{7}\right]$

$\boxed{\text{y21}}$ $\mathbb{M}_{1,1}^{(s)}(E) = \left[\frac{\sqrt{7}i}{14},\ \frac{\sqrt{7}i}{7},\ -\frac{3\sqrt{7}i}{14},\ -\frac{3\sqrt{7}i}{14},\ \frac{\sqrt{7}i}{14},\ \frac{\sqrt{7}i}{7}\right]$

$\boxed{\text{y22}}$ $\mathbb{M}_{1,2}^{(s)}(E) = \left[-\frac{5\sqrt{21}i}{42},\ \frac{2\sqrt{21}i}{21},\ \frac{\sqrt{21}i}{42},\ -\frac{\sqrt{21}i}{42},\ \frac{5\sqrt{21}i}{42},\ -\frac{2\sqrt{21}i}{21}\right]$

$\boxed{\text{y23}}$ $\mathbb{Q}^{(s)}_{2,1}(E,1) = \left[\frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7}, -\frac{3\sqrt{7}}{14}, -\frac{3\sqrt{7}}{14}, \frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7}\right]$

$\boxed{\text{y24}}$ $\mathbb{Q}^{(s)}_{2,2}(E,1) = \left[-\frac{5\sqrt{21}}{42}, \frac{2\sqrt{21}}{21}, \frac{\sqrt{21}}{42}, -\frac{\sqrt{21}}{42}, \frac{5\sqrt{21}}{42}, -\frac{2\sqrt{21}}{21}\right]$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	`A`	$\lvert s,\uparrow\rangle$, $\lvert s,\downarrow\rangle$, $\lvert p_x,\uparrow\rangle$, $\lvert p_x,\downarrow\rangle$, $\lvert p_y,\uparrow\rangle$, $\lvert p_y,\downarrow\rangle$, $\lvert p_z,\uparrow\rangle$, $\lvert p_z,\downarrow\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	`A`	`A`	`[1,2]`	`[s,p]`	`[s,p]`

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'A' (#1) site cluster (6c), 1

SL	position ($\boldsymbol{s}$)	mapping
1	[1.00000, 0.00000, 1.00000]	[1]
2	[0.00000, 1.00000, 1.00000]	[2]
3	[-1.00000,-1.00000, 1.00000]	[3]
4	[0.00000, 1.00000,-1.00000]	[4]
5	[1.00000, 0.00000,-1.00000]	[5]
6	[-1.00000,-1.00000,-1.00000]	[6]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 8: 1-th 'A'-'A' [1] (#1) bond cluster (6a@6c), D, $|\boldsymbol{v}|$= 1.73205 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-1.00000, 1.00000, 0.00000]	[0.50000, 0.50000, 1.00000]	[1]	(2,1)	(1,1)	[0,0,0]
2	[-1.00000,-2.00000, 0.00000]	[-0.50000, 0.00000, 1.00000]	[2]	(3,1)	(2,1)	[0,0,0]
3	[2.00000, 1.00000, 0.00000]	[0.00000,-0.50000, 1.00000]	[3]	(1,1)	(3,1)	[0,0,0]
4	[1.00000,-1.00000, 0.00000]	[0.50000, 0.50000,-1.00000]	[4]	(5,1)	(4,1)	[0,0,0]
5	[-2.00000,-1.00000, 0.00000]	[0.00000,-0.50000,-1.00000]	[5]	(6,1)	(5,1)	[0,0,0]
6	[1.00000, 2.00000, 0.00000]	[-0.50000, 0.00000,-1.00000]	[6]	(4,1)	(6,1)	[0,0,0]

Table 9: 2-th 'A'-'A' [1] (#2) bond cluster (3a@3b), ND, $|\boldsymbol{v}|$= 2.0 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000, 0.00000,-2.00000]	[1.00000, 0.00000, 0.00000]	[1,-5]	(5,1)	(1,1)	[0,0,0]
2	[0.00000, 0.00000,-2.00000]	[0.00000, 1.00000, 0.00000]	[2,-4]	(4,1)	(2,1)	[0,0,0]
3	[0.00000, 0.00000,-2.00000]	[-1.00000,-1.00000, 0.00000]	[3,-6]	(6,1)	(3,1)	[0,0,0]